

Fundamentals of Ecology and Evolutionary Biology

BIOS 20153 Lecture Notes

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1 Introduction

These are the lecture notes for **BIOS 20153 Fundamentals of Ecology and Evolutionary Biology**, Academic Year 2026-27.

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1.1 Ecology

Ecology is the study of the **distribution and abundance of organisms**.

It considers organisms at the individual, population, community, ecosystem, and biosphere levels. In this class, we are going to focus on the population and community levels. Ecology overlaps with the closely related sciences of biogeography, evolutionary biology, genetics, ethology, and natural history.

1 Introduction

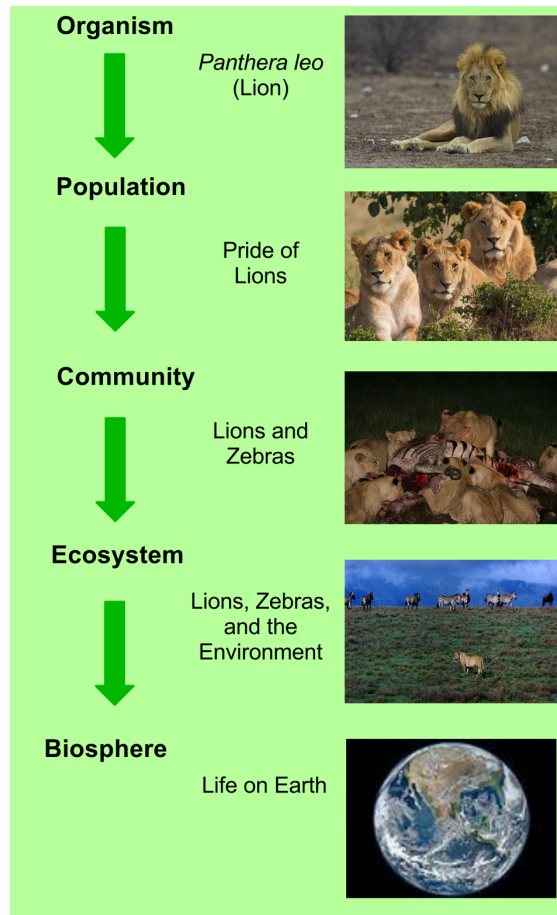


Figure 1.1: Levels of biological organization relevant to ecology. Image: [Wikimedia Commons](#), CC BY-SA 3.0.

1.2 Approach

Many factors determine the abundance and distribution of the organisms of a species, including:

- The number of individuals in the population
- The distribution of their age, sex, and size
- The availability of resources
- The presence of other species, including predators, prey, parasites, etc.
- The environmental conditions

Keeping track of these factors, and determine how the abundance of a species will change in time is very hard. Historically, the best strategy has been to take a **quantitative approach**, and write **mathematical models of population dynamics**.

Therefore, you will see a fair number of **equations** in this class. The prerequisites are single-variable calculus (and possibly some multivariate calculus)—all the rest will be explained in detail. Importantly:

- you will **not** have to perform extensive calculations on the final, but
- you will need to be able to **interpret equations that are provided**

1.3 Topics

1. Unbounded growth
 - a. Demography, age structure
 - b. Endangered species
2. Density-dependent growth
 - a. Logistic growth
 - b. Discrete generations & cycles

1 Introduction

- c. Fisheries management

3. Pathogens

- a. Consequences of vaccination
- b. Effects of diseases on animal populations

4. Population Cycles

- a. Predator-prey
- b. Host-pathogen

2 Unbounded growth

2.1 Overview

2.2 Historical models: geometric growth

Maybe the first example showing how growth can rapidly lead to large populations can be found in the *Liber abaci* (1202) by Leonardo Da Pisa, known as Fibonacci.

A certain man had one pair of rabbits together in a certain enclosed place, and one wishes to know how many are created from the pair in one year when it is the nature of them in a single month to bear another pair, and in the second month those born to bear also.

What is a bit unclear from the description: each pair of rabbits matures in one month, and once mature it produces a new pair of immature rabbits per month. Moreover, rabbits do not die! Then, calling R_t the number of rabbits at month t we obtain the famous Fibonacci's sequence:

$$1, 1, 2, 3, 5, 8, 13, \dots$$

By defining $R_1 = 1$ and $R_2 = 1$, we can write this implicitly as:

$$R_t = R_{t-1} + R_{t-2} \quad t \geq 3$$

2 Unbounded growth

or solve the recursion to obtain:

$$R_t = \frac{\phi^n - \psi^n}{\sqrt{5}}$$

where we have defined the golden ratio $\phi = \frac{1}{2}(1+\sqrt{5})$ and its conjugate $\psi = \frac{1}{2}(1-\sqrt{5})$. Thus, after one year (12 months) we will have 144 rabbits, and after 3 years $R_{36} = 14,930,352$. Let's plot the growth of the population in time

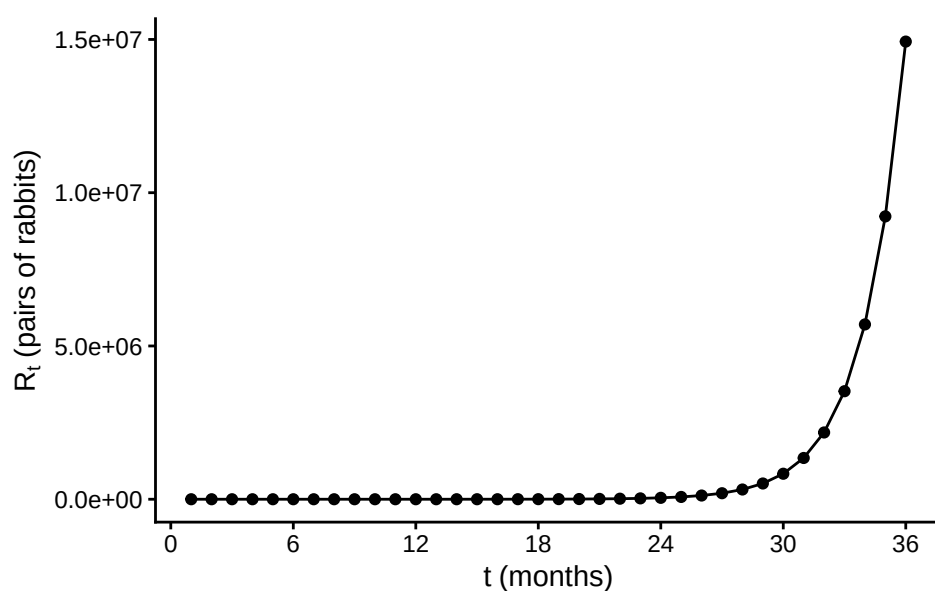


Figure 2.1: Fibonacci's rabbits: number of pairs R_t over 36 months.

Because the numbers grow so rapidly, it is convenient to plot $\log R_t$ vs t

You can see that after some jiggling, the curve in the logarithmic scale approaches a line; this is because, for $n \gg 1$ we have

2.2 Historical models: geometric growth

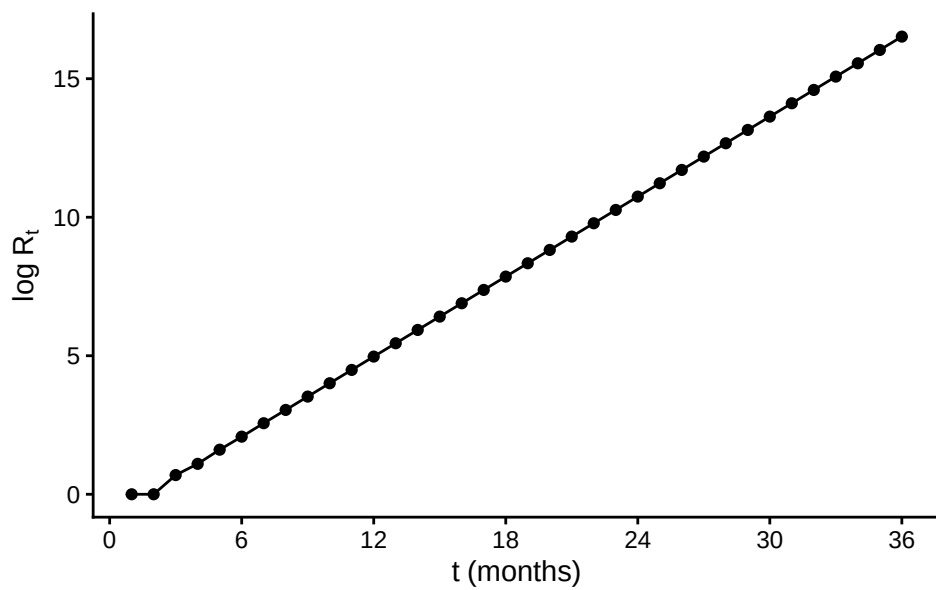


Figure 2.2: Log-scale plot of Fibonacci's rabbits: $\log R_t$ grows nearly linearly, reflecting exponential growth.

2 Unbounded growth

$$R_t \approx \phi R_{t-1}$$

and thus the number of rabbits will grow of approximately 62% per month. Soon, rabbits would take over the world!

i Rabbits in Australia

On Christmas day 1859, Thomas Austin released 24 breeding rabbits (*Oryctolagus cuniculus*) on his estate as game for shooting parties. The rabbits soon expanded across the continent, growing to concerning numbers and with devastating impacts on both agriculture and wildlife.

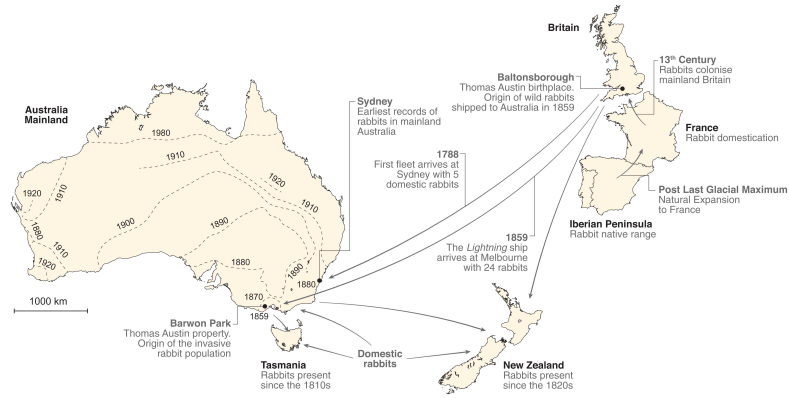


Figure 2.3: Rabbit expansion in Australia. Figure 1 from Alves et al. (2022).

Estimates vary widely (quantitative ecology was born in the late 1920s, and statistical ecology even later), but by 1910 there were about 1 billion rabbits roaming the continent. A recent genetic analysis (Alves et al. (2022)) has shown that they all descended from the original rabbits from Thomas Austin!

2.2 Historical models: geometric growth

In 1748, Leonard Euler published *Introductio in analysin infinitorum*, and included the following example:

If we assume that the human race was re-established by six individuals after the Flood—let us say, over a period of two hundred years—and that the population grew to one million, the question arises as to the rate at which the number of people must have increased annually.

He started from the equation:

$$P_{t+1} = (1 + x)P_t$$

with x the annual increase of the population. Iterating:

$$\begin{aligned}P_1 &= (1 + x)P_0 \\P_2 &= (1 + x)P_1 = (1 + x)(1 + x)P_0 = (1 + x)^2P_0 \\P_3 &= (1 + x)P_2 = (1 + x)^3P_0 \\&\dots\dots \\P_n &= (1 + x)P_{n-1} = (1 + x)^nP_0\end{aligned}$$

Then one needs to solve (note—we write $\log$ as the natural logarithm):

2 Unbounded growth

$$\begin{aligned}P_{200} &= (1+x)^{200}P_0 \\10^6 &= (1+x)^{200}6 \\ \frac{10^6}{6} &= (1+x)^{200} \\ \log \frac{10^6}{6} &= 200 \log(1+x) \\ 12.02 &\approx 200 \log(1+x) \\ 0.06 &\approx \log(1+x) \\ 1.062 &\approx 1+x \\ x &\approx 0.062\end{aligned}$$

He found the number reasonable, but mused:

But if, by the same rate of increase, the human population had grown over a period of 400 years, it would have risen to a number—16,666,666,666—which the entire earth would have been wholly inadequate to sustain.

We will return to this thought in the next chapter.

Geometric growth

These are all cases of **geometric growth**, in which the population in the next period is given by:

$$P_{t+1} = (1+x)P_t$$

If $x > 0$, the population will expand; if $x < 0$ it will contract; if it is exactly $x = 0$ it will remain constant. When the population is expanding, it will expand very rapidly; when it contracts, it will rapidly approach zero.

2.2 Historical models: geometric growth

The explosive nature of growth and decline has important consequences for natural populations, the spread of diseases, and even human society. Thomas Malthus, in the abrasive *An Essay on the Principle of Population* (Malthus (1798)) wrote that

Population, when unchecked, increases in a geometrical ratio. Subsistence increases only in an arithmetical ratio. A slight acquaintance with numbers will shew the immensity of the first power in comparison of the second.

He concluded that, as a result, population growth naturally outpaces the food supply, inevitably leading to poverty, famine, and population reduction unless checked by moral restraint, disease, war, or starvation. These bleak conclusions attracted praise and harsh critique, making this slim treatise one of the most influential books of the last few centuries.

Among the critics, Marx and Engels were among the most vitriolic. Engels described Malthus's thesis as (Engels (1844)):

the crudest, most barbarous theory that ever existed, a system of despair which struck down all those beautiful phrases about love thy neighbour and world citizenship

In the realm of biology, the *struggle for existence* implicit in the competition for scarce resources inspired the founders of evolution and provided a mechanism for natural selection. Darwin wrote in *The Origin of Species* (Darwin (1859)):

This is the doctrine of Malthus, applied with manifold force to the animal and vegetable kingdoms, for in this case there can be no artificial increase of food, and no prudential restraint from marriage

and in his autobiography:

2 Unbounded growth

In October 1838 [...] I happened to read for amusement ‘Malthus on Population’ [...] it at once struck me that under these circumstances favourable variations would tend to be preserved, and unfavourable ones to be destroyed. The result of this would be the formation of new species.

Similarly, Wallace wrote in his autobiography (Wallace (2016)):

But perhaps the most important book I read was Malthus’s Principles of Population [...] It was the first great work I had yet read treating of any of the problems of philosophical biology, and its main principles remained with me as a permanent possession, and twenty years later gave me the long-sought clue to the effective agent in the evolution of organic species

2.3 Exponential growth

The geometric growth model is well suited for populations where reproduction occurs in discrete time steps (such as seasonally breeding plants or animals). However, many populations, including humans and bacteria, reproduce continuously. To account for this, we can extend the discrete model into continuous time.

i Are births uniformly distributed throughout the year?

The exponential growth model assumes a constant per-capita birth rate. But are births actually spread evenly across the calendar? The figure below shows the average number of births per calendar day in the United States, based on Social Security Administration records for 2000–2014.

2.3 Exponential growth

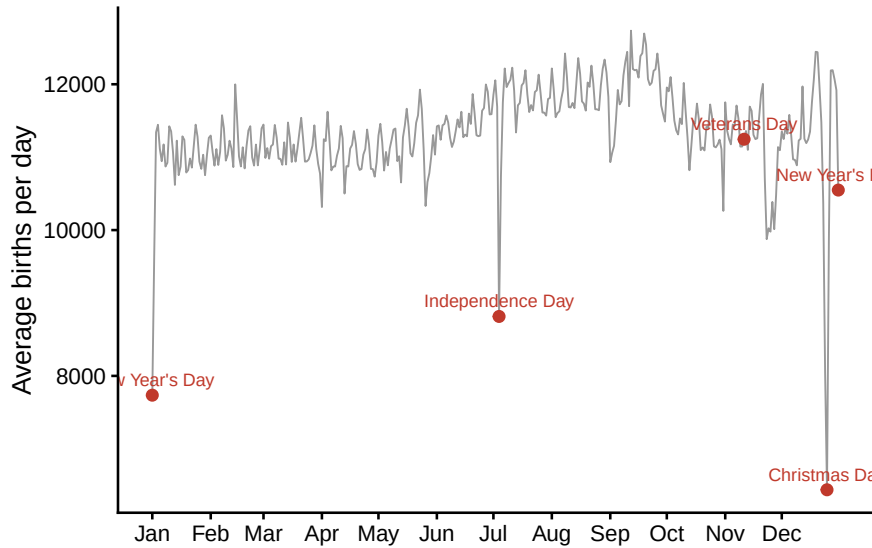


Figure 2.4: Average daily births in the US (SSA data, 2000–2014). Red points mark major public holidays.

Notice the sharp dips on Christmas Day, New Year's Day, and Independence Day. Why might that be? And what does it imply about the assumption of a constant birth rate in our model?

Suppose a population has an annual per-capita growth rate r . If reproduction occurs not just once a year, but is split into k discrete reproductive periods per year, the growth rate per interval is $\frac{r}{k}$. The population size at time t years becomes:

$$N_t = N_0 \left(1 + \frac{r}{k}\right)^{kt}$$

As reproduction becomes uninterrupted, the interval between births shrinks to zero ($k \rightarrow \infty$). In this continuous-time limit, we use the

2 Unbounded growth

definition of Euler's number (e):

$$e = \lim_{k \rightarrow \infty} \left(1 + \frac{1}{k}\right)^k \approx 2.71828$$

Taking the limit as $k \rightarrow \infty$ transforms the discrete compounding model into the continuous exponential growth equation:

$$N(t) = N(0)e^{rt}$$

Here, $N(t)$ denotes population size at any time $t \geq 0$, whereas the discrete model only defined population size at integer steps ($t = 0, 1, 2, \dots$).

! Key Distinction

In discrete growth, population size N_t is only defined at specific intervals ($t = 0, 1, 2, \dots$). In continuous growth, $N(t)$ is defined for **all** $t \geq 0$.

To find how fast the population is growing at any instant, we take the derivative of $N(t)$ with respect to time:

$$\frac{dN(t)}{dt} = \frac{d}{dt} [N(0)e^{rt}]$$

Using the exponential derivative rule ($\frac{d}{dt}e^{rt} = re^{rt}$):

$$\frac{dN(t)}{dt} = r \cdot \underbrace{N(0)e^{rt}}_{N(t)} = rN(t)$$

2.3 Exponential growth

Biological Meaning

This reveals that the instantaneous growth rate ($\frac{dN}{dt}$) is simply the per-capita rate r multiplied by the current population size $N(t)$. The interpretation of r is the difference between birth and death rates per individual. If $r > 0$, the population grows; if $r < 0$, it declines; if $r = 0$, it remains constant.

3 Density dependence

3.1 Overview

4 Pathogens

4.1 Overview

5 Predator-prey dynamics

5.1 Overview

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